

DAY-2 Assignment

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1. Create Database

```
create database InsuranceDB;
```

2. Creating Tables

```
create table Customers(  
CustomerID int,  
FirstName varchar(20),  
LastName varchar(25),  
DateOfBirth date,  
Phone varchar(10),  
Email varchar(20),  
Primary key(CustomerID)  
)
```

```
create table Policies(  
PolicyID int,  
PolicyName varchar(20),  
PolicyType varchar(30),  
PremiumAmount money,  
DurationYears int  
Primary key(PolicyID)  
)
```

```
create table Agents(  
AgentID int,  
AgentName varchar(20),  
Phone varchar(10),  
City varchar(15),  
Primary key(AgentID)  
)
```

```
create table PolicyAssignments(  
AssignmentID int,  
CustomerID int,  
PolicyID int,  
AgentID int,  
StartDate date,  
EndDate date,  
primary key(AssignmentID),  
foreign key(PolicyID) References Policies(PolicyID),  
foreign key(CustomerID) References Customers(CustomerID),  
foreign key(AgentID) References Agents(AgentID)  
)
```

```
create table Claims(  
ClaimID int,  
AssignmentID int,  
ClaimDate date,
```

```

ClaimAmount varchar(10),
ClaimStatus varchar(10)
Primary key(ClaimID),
foreign key(AssignmentID) references PolicyAssignments(AssignmentID)
)

```

```

create table Payments(
PaymentID int, PolicyID int, PaymentDate date, Amount money,
primary key(PaymentID),
foreign key(PolicyID) references Policies(PolicyID));

```

```

Alter table Claims
alter column ClaimAmount money;

```

3. Inserting data into Tables

```

insert into Customers
values (1, 'John', 'darwin', '1985-12-02', '9999999999', 'johnd@gmail.com');

```

```

insert into Customers
values(2, 'Anita', 'Sharma', '1990-06-15', '8888888888', 'anita@gmail.com'),
(3, 'Rahul', 'Verma', '1988-03-22', '7777777777', 'rahul@gmail.com'),
(4, 'Sneha', 'Patel', '1995-11-10', '6666666666', 'sneha@gmail.com');

```

```

insert into Policies
values
(1, 'Home Secure', 'Home Insurance', 25000, 10),
(2, 'Life Shield', 'Life Insurance', 18000, 20),
(3, 'Health Plus', 'Health Insurance', 12000, 5),
(4, 'Vehicle Safe', 'Vehicle Insurance', 15000, 3);

```

```

insert into Agents
values(1, 'Ramesh Kumar', '9123456789', 'Hyderabad'),
(2, 'Sita Rao', '9234567890', 'Bangalore'),
(3, 'Amit Singh', '9345678901', 'Delhi');

```

```

insert into PolicyAssignments
values(1,1,1,1, '2022-01-01', '2032-01-01'),
(2,2,2,2, '2021-06-15', '2041-06-15'),
(3,3,3,3, '2023-03-10', '2028-03-10'),
(4,4,4,1, '2024-02-01', '2027-02-01');

```

```

insert into Claims
values(1,1, '2023-05-20', 50000, 'Approved'),
(2,2, '2022-08-10', 100000, 'Pending'),
(3,3, '2024-01-05', 25000, 'Rejected');

```

4.1

select * from Customers;

Results		Messages				
	CustomerID	FirstName	LastName	DateOfBirth	Phone	Email
1	1	John	darwin	1985-12-02	9999999999	johnd@gmail.com
2	2	Anita	Sharma	1990-06-15	8888888888	anita@gmail.com
3	3	Rahul	Verma	1988-03-22	7777777777	rahul@gmail.com
4	4	Sneha	Patel	1995-11-10	6666666666	sneha@gmail.com

4.2

select CustomerID,PolicyID,StartDate,EndDate from PolicyAssignments;

	CustomerID	PolicyID	StartDate	EndDate
1	1	1	2022-01-01	2032-01-01
2	2	2	2021-06-15	2041-06-15
3	3	3	2023-03-10	2028-03-10
4	4	4	2024-02-01	2027-02-01

4.3

select * from Policies where PolicyType='Health Insurance';

	PolicyID	PolicyName	PolicyType	PremiumAmount	DurationYears
1	3	Health Plus	Health Insurance	12000.00	5
2	5	Health Basic	Health Insurance	11000.00	1

4.4

select * from Policies where PremiumAmount>10000 and DurationYears=1;

	PolicyID	PolicyName	PolicyType	PremiumAmount	DurationYears
1	5	Health Basic	Health Insurance	11000.00	1

4.5

select Distinct(City) from Agents;

	City
1	Bangalore
2	Delhi
3	Hyderabad

4.6

select * from Policies where PolicyType='Home Insurance' or PolicyType='Life Insurance' or PolicyType='Motor Insurance';

	PolicyID	PolicyName	PolicyType	PremiumAmount	DurationYears
1	1	Home Secure	Home Insurance	25000.00	10
2	2	Life Shield	Life Insurance	18000.00	20
3	6	Motor Secure	Motor Insurance	10000.00	1

4.7

select * from Policies where PolicyType in ('Home Insurance','Motor Insurance','Life Insurance');

	PolicyID	PolicyName	PolicyType	PremiumAmount	DurationYears
1	1	Home Secure	Home Insurance	25000.00	10
2	2	Life Shield	Life Insurance	18000.00	20
3	6	Motor Secure	Motor Insurance	10000.00	1

4.8

select * from Customers where DateOfBirth>='2001-01-01' and DateOfBirth<='2022-12-31';

	CustomerID	FirstName	LastName	DateOfBirth	Phone	Email
1	5	Aarav	Kumar	2002-03-15	9000000001	aarav@gmail.com
2	6	Meera	Shah	2005-07-22	9000000002	meera@gmail.com
3	7	Rohan	Patel	2010-01-10	9000000003	rohan@gmail.com
4	8	Isha	Reddy	2021-02-05	9000000004	isha@gmail.com

4.9

select * from Customers where DateOfBirth between '2001-01-01' and '2022-12-31';

	CustomerID	FirstName	LastName	DateOfBirth	Phone	Email
1	5	Aarav	Kumar	2002-03-15	9000000001	aarav@gmail.com
2	6	Meera	Shah	2005-07-22	9000000002	meera@gmail.com
3	7	Rohan	Patel	2010-01-10	9000000003	rohan@gmail.com
4	8	Isha	Reddy	2021-02-05	9000000004	isha@gmail.com

4.10

select * from Claims where ClaimStatus='Rejected';

	ClaimID	AssignmentID	ClaimDate	ClaimAmount	ClaimStatus
1	3	3	2024-01-05	25000.00	Rejected

4.11

select * from Agents where City like '_a%';

	AgentID	AgentName	Phone	City
1	2	Sita Rao	9234567890	Bangalore

4.12

select max(ClaimAmount) as highest,min(ClaimAmount) as lowest from Claims;

	highest	lowest
1	100000.00	25000.00

4.13

select TOP 1 * from Claims

order by ClaimDate desc;

	ClaimID	AssignmentID	ClaimDate	ClaimAmount	ClaimStatus
1	3	3	2024-01-05	25000.00	Rejected

4.14

update Policies set PremiumAmount=PremiumAmount*1.1 where PolicyType='Health Insurance';

select * from Policies where PolicyType='Health Insurance';

	PolicyID	PolicyName	PolicyType	PremiumAmount	DurationYears
1	3	Health Plus	Health Insurance	13200.00	5
2	5	Health Basic	Health Insurance	12100.00	1

4.15

delete from PolicyAssignments

where EndDate<CAST(getdate() as date);

	AssignmentID	CustomerID	PolicyID	AgentID	StartDate	EndDate
1	1	1	1	1	2022-01-01	2032-01-01
2	2	2	2	2	2021-06-15	2041-06-15
3	3	3	3	3	2023-03-10	2028-03-10
4	4	4	4	1	2024-02-01	2027-02-01
5	11	2	2	2	2023-01-01	2026-12-31

4.16

select count(*) as count from Claims where ClaimStatus='Rejected';

Results Messages	
	count
1	1

4.17

**select Policyid, Policyname, PremiumAmount,
PremiumAmount*0.06 as localTaxes,
PremiumAmount*1.06 as PremiumAmountWithTax,
(premiumamount*1.06)/12 as MonthlyPremiumAmount
from Policies;**

Results

Messages

	Policyid	Policyname	PremiumAmount	localTaxes	PremiumAmountWithTax	MonthlyPremiumAmount
1	1	Home Secure	25000.00	1500.000000	26500.000000	2208.333333333
2	2	Life Shield	18000.00	1080.000000	19080.000000	1590.000000000
3	3	Health Plus	13200.00	792.000000	13992.000000	1166.000000000
4	4	Vehicle Safe	15000.00	900.000000	15900.000000	1325.000000000
5	5	Health Basic	12100.00	726.000000	12826.000000	1068.833333333
6	6	Motor Secure	10000.00	600.000000	10600.000000	883.333333333

4.18

**alter table Customers add Address varchar(50), City varchar(30);
update Customers set Address='Mg Road', City='Hyderabad' where CustomerId=1;
select * from Customers;**

Results Messages

	CustomerID	FirstName	LastName	DateOfBirth	Phone	Email	Address	City
1	1	John	darwin	1985-12-02	9999999999	johnd@gmail.com	Mg Road	Hyderabad
2	2	Anita	Sharma	1990-06-15	8888888888	anita@gmail.com	NULL	NULL
3	3	Rahul	Verma	1988-03-22	7777777777	rahul@gmail.com	NULL	NULL

4.19

alter table Agents add DevOfld int;

4.20

**alter table Agents
add constraint FkAgentsDevOfld
foreign key(DevOfld) references Agents(AgentId);
select TOP (2) * from Agents;**

Results		Messages				
	AgentID	AgentName	Phone	City	DevOfid	
1	1	Ramesh Kumar	9123456789	Hyderabad	NULL	
2	2	Sita Rao	9234567890	Bangalore	NULL	

5.1

```
select p.PolicyId, p.PolicyName, p.PolicyType, p.PremiumAmount
from PolicyAssignments pa
join Policies p on p.PolicyId=pa.PolicyId
where pa.CustomerId=5;
```

Results		Messages			
	PolicyId	PolicyName	PolicyType	PremiumAmount	
1	1	Home Secure	Home Insurance	25000.00	

5.2

```
select c.CustomerId, c.FirstName, p.PolicyName
from Customers c
join PolicyAssignments pa on pa.CustomerId=c.CustomerId
join Policies p on p.PolicyId=pa.PolicyId;
```

Results		Messages	
	CustomerId	FirstName	PolicyName
1	1	John	Home Secure
2	2	Anita	Life Shield
3	3	Rahul	Health Plus
4	4	Sneha	Vehicle Safe
5	2	Anita	Life Shield
6	5	Aarav	Home Secure

5.3

```
select c.FirstName, cl.ClaimId, cl.ClaimAmount, cl.ClaimStatus
from Claims cl
join PolicyAssignments pa on pa.AssignmentId=cl.AssignmentId
join Customers c on c.CustomerId=pa.CustomerId;
```

Results Messages				
	FirstName	ClaimId	ClaimAmount	ClaimStatus
1	John	1	50000.00	Approved
2	Anita	2	100000.00	Pending
3	Rahul	3	25000.00	Rejected

5.4

```
select c.FirstName, p.PolicyName, a.AgentName, pa.StartDate, pa.EndDate
from PolicyAssignments pa
join Customers c on c.CustomerId=pa.CustomerId
join Policies p on p.PolicyId=pa.PolicyId
join Agents a on a.AgentId=pa.AgentId;
```

Results Messages					
	FirstName	PolicyName	AgentName	StartDate	EndDate
1	John	Home Secure	Ramesh Kumar	2022-01-01	2032-01-01
2	Anita	Life Shield	Sita Rao	2021-06-15	2041-06-15
3	Rahul	Health Plus	Amit Singh	2023-03-10	2028-03-10
4	Sneha	Vehicle Safe	Ramesh Kumar	2024-02-01	2027-02-01
5	Anita	Life Shield	Sita Rao	2023-01-01	2026-12-31
6	Aarav	Home Secure	Ramesh Kumar	2024-01-01	2026-01-01

5.5

```
select c.FirstName, p.PolicyName, cl.ClaimAmount, cl.ClaimStatus, cl.ClaimDate
from Claims cl
join PolicyAssignments pa on pa.AssignmentId=cl.AssignmentId
join Customers c on c.CustomerId=pa.CustomerId
join Policies p on p.PolicyId=pa.PolicyId;
```

Results Messages					
	FirstName	PolicyName	ClaimAmount	ClaimStatus	ClaimDate
1	John	Home Secure	50000.00	Approved	2023-05-20
2	Anita	Life Shield	100000.00	Pending	2022-08-10
3	Rahul	Health Plus	25000.00	Rejected	2024-01-05

5.6

```
select c.CustomerId, c.FirstName, p.PolicyName
from Customers c
left join PolicyAssignments pa on pa.CustomerId=c.CustomerId
left join Policies p on p.PolicyId=pa.PolicyId;
```

	CustomerId	FirstName	PolicyName
1	1	John	Home Secure
2	2	Anita	Life Shield
3	2	Anita	Life Shield
4	3	Rahul	Health Plus
5	4	Sneha	Vehicle Safe
6	5	Aarav	Home Secure
7	6	Meera	NULL
8	7	Rohan	NULL
9	8	Isha	NULL

5.7

```
select c.CustomerId, c.FirstName from Customers c
where c.CustomerId not in (
select pa.CustomerId from PolicyAssignments pa
join Claims cl on cl.AssignmentId=pa.AssignmentId
);
```

	CustomerId	FirstName
1	4	Sneha
2	5	Aarav
3	6	Meera
4	7	Rohan
5	8	Isha

5.8

```
select c.FirstName, sum(cl.ClaimAmount) as TotalClaimAmount
from Claims cl
join PolicyAssignments pa on pa.AssignmentId=cl.AssignmentId
join Customers c on c.CustomerId=pa.CustomerId
group by c.FirstName;
```

Results Messages		
	FirstName	TotalClaimAmount
1	Anita	100000.00
2	John	50000.00
3	Rahul	25000.00

5.9

```
select c.FirstName, sum(cl.ClaimAmount) as TotalClaimAmount
from Claims cl
join PolicyAssignments pa on pa.AssignmentId=cl.AssignmentId
join Customers c on c.CustomerId=pa.CustomerId
group by c.FirstName
having sum(cl.ClaimAmount)>50000;
```

Results Messages		
	FirstName	TotalClaimAmount
1	Anita	100000.00

5.10

```
select a.AgentName, count(pa.PolicyId) as PolicyCount
from Agents a
left join PolicyAssignments pa on pa.AgentId=a.AgentId
group by a.AgentName;
```

Results Messages		
	AgentName	PolicyCount
1	Amit Singh	1
2	Ramesh Kumar	3
3	Sita Rao	2
4	Suresh	0